

INTELLIGENT VIBRATION ANALYSIS FOR DRONE MALFUNCTION DETECTION

MINI PROJECT REPORT

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DECLARATION

We undersigned hereby declare that the project report “**Intelligent Vibration Analysis for Drone Malfunction Detection**” , submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a Bonafide work done by us under supervision of Ms. Asha George. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled “**Intelligent Vibration Analysis for Drone Malfunction Detection**” submitted by **ADHIL B, HAMNA MOHAMMED RAFFI, GREESHMA S KUMAR, ABDULLA ANAS** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of degree of Bachelor of Technology in **COMPUTER SCIENCE AND ENGINEERING** is a Bonafide record of the project work carried out by them under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

Drones have become essential tools in modern industries such as surveillance, delivery, agriculture, and disaster management. As their usage increases, ensuring their safety and reliability has become a major concern. Mechanical faults in drones can lead to serious accidents, especially during flight. One of the early signs of such faults is abnormal vibration. This project explores a smart approach to detect potential drone malfunctions through vibration analysis. By monitoring the vibrations produced by key drone components—such as motors and propellers—it's possible to identify early warning signs of failure. To make this process more efficient and accurate, we apply machine learning techniques that can analyze the data and predict the condition of the drone in real time. Our goal is to develop a system that not only detects faults accurately but also makes the information easily accessible to users through a web application. This helps drone operators take timely action, reducing risks and improving maintenance. The system uses real-time vibration data, classifies it based on severity, and provides an immediate fault status. It is designed to be user-friendly, making it suitable even for non-technical users. A key highlight of this project is the shift from traditional manual fault detection to an automated, intelligent solution. In the future, the system can be expanded to outdoor environments, where factors like wind and terrain may affect drone performance. There is also potential for integrating the solution with onboard sensors and advanced algorithms like deep learning for improved accuracy. Overall, this project aims to enhance drone safety by combining sensor data and intelligent analysis, offering a practical solution for real-world applications.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Drones have become essential in diverse fields such as surveillance, disaster management, agriculture, and logistics. However, mechanical malfunctions, particularly in motors, propellers, or structural components, pose serious risks during operation. Traditional fault detection techniques are often reactive and limited in scope. Hence, there is a growing need for intelligent, data-driven approaches that can detect faults early. This project focuses on leveraging vibration analysis—a reliable indicator of mechanical health—combined with AI techniques to build a smart, proactive fault detection system for drones.

Drones, also known as unmanned aerial vehicles (UAVs), have emerged as transformative tools across various sectors due to their versatility and efficiency. They are widely used in surveillance for monitoring and security purposes, in agriculture for crop monitoring and spraying, in logistics for delivering goods, and in disaster management for search and rescue operations, damage assessment, and relief delivery. Despite their growing adoption and usefulness, drones face significant operational challenges. These challenges often stem from mechanical failures, motor and propeller issues, structural imbalances, or adverse environmental conditions such as wind, rain, and temperature extremes. Such factors can lead to performance degradation, mission failure, or even catastrophic crashes, posing safety risks and financial losses.

Traditional fault detection mechanisms in drones typically rely on post-failure diagnostics or scheduled maintenance, which are often reactive rather than proactive. This reactive approach limits the ability to prevent failures before they occur. Hence, there is a pressing need for intelligent and proactive systems that can detect faults early and provide timely alerts or corrective actions. One of the most effective indicators of mechanical health in drones is vibration.

1.2 OVERVIEW

The project, titled “Intelligent Vibration Analysis for Drone Malfunction Detection,” aims to develop a system that monitors vibration data in real-time and uses machine learning models to classify drone conditions as safe, partially safe, or unsafe. By analyzing the vibration signatures

from critical drone parts like motors and arms, the system can detect anomalies indicating mechanical faults. The project includes fault simulation, data acquisition, model training using Random Forest, and deployment on a web platform for real-time user interaction and monitoring.

1.3 PROBLEM STATEMENT

Existing drone malfunction detection systems rely heavily on visual inspection or basic sensor feedback, which often fail to identify internal mechanical issues early. These systems tend to be reactive, detecting faults only after damage has occurred. Moreover, current vibration-based methods lack intelligent analysis capabilities and real-time responsiveness. This gap leads to increased maintenance costs, reduced operational safety, and potential mission failures. There is a clear need for a smarter solution that enables early detection and classification of drone malfunctions using advanced vibration analysis and machine learning.

1.3.1 Limitations

i. **Limited Real-world Testing:**

The system is primarily tested on simulated data (MATLAB, Kaggle). It may not fully capture the complexity of real flight conditions like wind, vibrations from terrain, or external disturbances.

ii. **Sensor Dependency:**

The system relies on external vibration sensors, which may add weight or affect the drone's performance. Integration with built-in sensors is yet to be explored.

iii. **Scope of Faults:**

Only mechanical faults related to vibration patterns are addressed. Electrical, battery-related, or software-level faults are not considered.

iv. **Fixed Thresholds for Vibration Amplitudes:**

The classification system uses predefined amplitude ranges, which may not generalize well across different drone models or environments.

v. **AI Model Limitation:**

While Random Forest works well for classification, it may not handle complex, high-dimensional datasets as effectively as deep learning models in the future.

1.3.2 Objectives

- To develop a smart, AI-based system for detecting and classifying drone malfunctions.
- To analyze vibration data from drone components for identifying early signs of mechanical issues.
- To enhance the reliability, performance, and safety of drone operations.
- To provide early warnings of potential faults, enabling timely interventions and reduced downtime.
- To apply machine learning techniques for accurate fault classification.
- To contribute to the broader fields of predictive maintenance and intelligent fault detection in aerial robotics.
- To address a critical need in modern drone technology related to real-time diagnostics.

1.4 INDUSTRY RELEVANCE

- **Widespread Drone Adoption:** Drones are increasingly used in sectors like logistics, agriculture, surveillance, and disaster management, highlighting the need for reliable and safe operation.
- **Increasing Safety Regulations:** As drones are deployed in more sensitive or crowded environments, there is a higher demand for safety-focused technologies and early fault detection systems.
- **Proven Value of Vibration Analysis:** Vibration monitoring is already widely used in industrial machines, and applying it to drones shows the potential of extending proven methods to aerial systems.
- **Demand for Smart Diagnostic Systems:** Industries are seeking intelligent, automated solutions that reduce human dependency and allow real-time monitoring and alerts.

1.5 PROPOSED SYSTEM

The proposed system is designed to detect and classify mechanical faults in drones by analyzing their vibration patterns. A vibration sensor is mounted on the drone's body to monitor its mechanical behavior during operation. These vibrations are indicative of the health of components

such as motors, arms, and propellers.

The collected vibration data is processed and analyzed using a machine learning model, specifically the Random Forest algorithm, which has been chosen for its better accuracy compared to traditional methods like Linear Regression. The system classifies the drone's condition into three

To train and validate the model, datasets were sourced from MATLAB and Kaggle, and various fault scenarios were simulated. The trained model is then integrated into a web-based interface that allows users to view the health status of the drone in real-time. This setup provides an efficient and proactive way to detect faults early, thereby improving the overall safety, reliability, and longevity of drone operations.

CHAPTER 2

REVIEW OF LITERATURE

2.1 VIBRATION DETECTION (Memari, 2024)

"Paper on the Advancements in Wind Turbine Blade Inspection Integrating Drone and Deep Learning Technologies for Enhanced Defect Detection" provides a comprehensive overview of how drones and deep learning are revolutionizing the inspection and maintenance of wind turbine blades (Memari, 2024). Traditionally, inspecting turbine blades has been a risky, labor-intensive, and time-consuming process, often involving human workers climbing turbines or using costly equipment like cranes. The paper highlights how the integration of drone technology has dramatically improved this process. Drones provide a safer, faster, and more cost-effective alternative, capturing high-resolution images and videos of turbine blades from various angles without interrupting operations. Alongside drones, the rise of deep learning techniques, especially Convolutional Neural Networks (CNNs) and other AI models, has greatly enhanced the ability to automatically detect, classify, and quantify defects such as cracks, erosion, and lightning strikes on the blades. The document discusses various deep learning architectures and their performance in real-world defect detection tasks. It emphasizes the importance of large, annotated datasets for training accurate models and mentions several data augmentation strategies to address the limited availability of defect data. The review also touches on challenges facing this integrated approach, such as dealing with environmental noise (like varying lighting conditions and blade surface dirt), managing large volumes of inspection data, and ensuring the generalizability of AI models across different turbine types and manufacturers. Techniques like transfer learning, unsupervised learning, and semi-supervised learning are explored as potential solutions. Moreover, the paper outlines recent advancements, including the use of 3D modeling, thermal imaging, and multimodal data fusion to improve defect detection accuracy. It notes the growing trend toward fully automated inspection systems, where drones autonomously fly predetermined paths and AI systems analyze the data in real time. In its conclusion, the document stresses the promising future of drone and AI-powered inspections but cautions that further research is needed to overcome current limitations. Key areas for future work include improving real-time defect detection, enhancing model robustness under diverse conditions, and developing standard protocols for automated inspections. Overall, the document paints an optimistic picture of how drones and deep learning are transforming wind turbine maintenance, contributing to more efficient renewable energy generation while reducing operational costs and human risk.

2.2 DRONE NAVIGATION (Arshad, 2022)

The paper titled "Drone Navigation Using Region and Edge Exploitation-Based Deep CNN" presents a novel approach for improving drone navigation using deep learning, particularly by enhancing how a drone interprets and moves through its environment (Arshad, 2022). The paper proposes a method that integrates region-based and edge-based information into a Deep Convolutional Neural Network (CNN) framework. Traditional drone navigation often relies either on GPS, manual remote control, or basic vision-based techniques, which can struggle with dynamic or complex environments. To overcome these challenges, the authors suggest that focusing on both regions (broad areas like pathways, obstacles, or open spaces) and edges (boundaries of objects or features) allows the drone to understand its surroundings more precisely and make better navigational decisions.

The proposed model consists of two primary components:

1. **Region Exploitation:** Identifying and understanding different regions in the drone's field of view, allowing it to recognize safe paths and avoid obstacles.
2. **Edge Exploitation:** Detecting and leveraging the edges or boundaries within an image to better define object outlines, ensuring more accurate localization and obstacle detection.

By combining these two types of visual features, the system can perform more reliable environment perception, even in visually cluttered or ambiguous settings. The authors designed a deep CNN architecture specifically optimized for real-time drone navigation tasks. They trained and evaluated their model on a custom dataset featuring various navigation scenarios with different obstacles, lighting conditions, and environmental complexities. The results show that their region-and-edge-enhanced deep CNN outperforms traditional CNN-based navigation systems in both accuracy and robustness, especially in environments with poor visibility or complex layouts. The paper also discusses experimental evaluations, comparing their approach to baseline models. Their method achieved higher success rates in navigation tasks, demonstrating better obstacle avoidance and path following abilities. They highlight that incorporating edge and region information reduces errors caused by misinterpretations of the scene, a common problem in previous models. In conclusion, the study emphasizes that exploiting detailed visual cues — specifically, regions and edges — allows drones to better "understand" their surroundings, making autonomous navigation safer and more efficient. The authors suggest that future work could

involve extending this approach with reinforcement learning or 3D environmental mapping for even more advanced autonomous navigation capabilities. Overall, the paper contributes a significant improvement in the field of vision-based drone navigation by combining sophisticated image analysis techniques with deep learning models.

2.3 PROTOTYPE IMPLEMENTATION (Chen, 2022)

The paper on "DroneTalk: An Internet-of-Things-Based Drone System for Last-Mile Drone Delivery" presents an innovative approach to solving challenges in last-mile delivery using Internet of Things (IoT) technology integrated with drone systems (Chen, 2022). The paper introduces DroneTalk, a system that enhances the reliability, safety, and efficiency of drone-based delivery services. Traditional last-mile delivery, which covers the final leg of a product's journey to the consumer, is often expensive, slow, and logistically complex. Drones offer a potential solution due to their speed and flexibility, but they also face challenges like communication failures, delivery verification, dynamic environment handling, and security risks. DroneTalk addresses these issues by embedding IoT capabilities directly into the drone system. It creates a communication bridge between drones, recipients, and the delivery infrastructure. Specifically, the system includes:

- IoT-Enabled Communication Modules that allow drones to interact with IoT devices at delivery locations, such as smart mailboxes or delivery pads.
- Authentication Mechanisms to ensure that deliveries are verified and secure, preventing package theft or mis delivery.
- Real-Time Monitoring of drone operations, providing data on location, package status, and environmental conditions.
- Intelligent Routing and Scheduling, utilizing IoT data to adapt routes dynamically in response to traffic, weather, or other obstacles.

The architecture of DroneTalk is described as modular and scalable, comprising a cloud platform for data management, edge devices for local processing, and drones equipped with IoT sensors and communication units. By using this layered approach, DroneTalk improves data efficiency and reduces response times, which are critical in dynamic last-mile environments. The paper also discusses prototype implementation and experimental validation. A small-scale testbed was developed, demonstrating that DroneTalk-enabled drones could successfully perform authenticated deliveries and maintain robust communication even in the presence of network disruptions. Their experiments showed improved delivery success rates and better adaptability compared to conventional drone delivery systems. In conclusion, the authors argue that DroneTalk represents a significant advancement toward making drone-based last-mile delivery systems practical and reliable. They highlight future work directions, including enhancing the

scalability of the system to support larger networks of drones and delivery points, integrating machine learning for better decision-making, and strengthening cybersecurity measures. Overall, the paper contributes to the growing field of smart logistics by showing how combining IoT and drone technologies can overcome traditional limitations and pave the way for more efficient, secure, and user-friendly delivery services.

2.4 FAULT DIAGNOSIS (Hussain, 2024)

The Paper on "Condition Monitoring and Fault Diagnosis of Wind Turbine: A Systematic Literature Review" provides a comprehensive survey of research developments focused on detecting and diagnosing faults in wind turbines to improve their reliability, efficiency, and maintenance strategies.

Wind turbines, being critical components of renewable energy systems, face operational challenges due to harsh environmental conditions, mechanical stresses, and system complexities(Hussain, 2024).To ensure their longevity and performance, Condition Monitoring (CM) and Fault Diagnosis (FD) techniques are essential. This paper systematically reviews existing methods, technologies, and research trends in this area. The review first categorizes CM and FD techniques into several groups:

- Signal-based approaches that use sensor data like vibration, temperature, or acoustic signals to monitor equipment health.
- Model-based methods that rely on mathematical or physical models to detect deviations indicating faults.
- Data-driven techniques, especially those leveraging machine learning and artificial intelligence, which have gained prominence for their ability to learn complex patterns without detailed physical modeling.The paper discusses key components of wind turbines—such as the gearbox, generator, blades, bearings, and tower structure—and the typical faults they experience, including gear tooth cracks, blade cracks, bearing wear, generator failure, and structural fatigue. For each component, the authors summarize the specific monitoring techniques applied and evaluate their effectiveness.A major focus is on the emergence of artificial intelligence (AI) and machine learning (ML) in recent research. Techniques such as neural networks, support vector machines, deep learning, and ensemble methods have significantly improved fault detection accuracy and early diagnosis, reducing downtime and maintenance costs. The integration of Internet of Things (IoT) devices and cloud computing for real-time monitoring is also highlighted as a growing trend. Furthermore, the paper identifies challenges in the field:
- The need for large, high-quality datasets for training AI models.

- Balancing model complexity with real-time monitoring requirements.
- Addressing the uncertainty and variability of operational environments.
- Developing standardized benchmarks for fair comparison between different methods.

The review concludes by suggesting future research directions, such as creating hybrid approaches combining signal-based and AI-based methods, improving fault prognosis (predicting future failures), and developing more robust systems capable of handling a wide variety of operational conditions.

In summary, this paper offers a detailed and structured overview of the state-of-the-art in wind turbine CM and FD, highlighting the transition from traditional techniques to intelligent, data-driven solutions aimed at ensuring sustainable and cost-effective wind energy generation.

2.5 VIBRATION ANALYSIS (Ghazali, 2022)

The paper on "Vibration-Based Fault Detection in Drone Using Artificial Intelligence" discusses a method for identifying mechanical faults in drones by analyzing vibration signals and applying Artificial Intelligence (AI) techniques. Drones, especially in commercial and industrial applications, are prone to mechanical issues like motor failures, propeller imbalances, and structural cracks due to continuous operation in varying environments (Ghazali, 2022). Early detection of these faults is critical to ensure flight safety, prevent crashes, and minimize maintenance costs. Traditional maintenance methods often rely on manual inspections, which can be inefficient and unable to predict failures early enough. This study proposes a vibration-based monitoring system where sensors collect vibration signals from critical drone components during operation. These signals carry essential information about the health status of the drone. However, manually interpreting this complex and massive data is impractical, so the authors leverage AI to automate fault detection.

The document outlines a complete framework:

- **Data Collection:** Vibration data is captured from drones under different operating conditions, including healthy states and various fault conditions.
- **Preprocessing:** Raw vibration signals are cleaned and transformed into formats suitable for AI analysis, often involving feature extraction (e.g., statistical features, frequency domain characteristics).
- **Model Training:** Machine learning algorithms, such as Support Vector Machines (SVM), Decision Trees, and deep learning models like Convolutional Neural Networks (CNNs), are trained to recognize patterns associated with different fault types.
- **Fault Classification:** Once trained, the models can classify incoming vibration signals into

categories like "healthy," "minor fault," or "critical fault."

The study emphasizes the advantages of using AI, particularly deep learning, which can automatically learn complex features from raw data without manual feature engineering. Deep models like CNNs have shown higher fault detection accuracy and robustness compared to traditional methods.

Experimental results demonstrate that the AI-based system can accurately detect faults with high precision and recall rates, significantly outperforming manual inspection and simple threshold-based methods. The system is capable of real-time monitoring, making it practical for deployment on operational drones. The paper also discusses challenges such as noise in vibration data, the need for large labeled datasets for effective model training, and potential generalization issues across different drone models. In conclusion, the document shows that vibration-based monitoring combined with AI offers a powerful, automated, and reliable solution for fault detection in drones, contributing to safer and more efficient drone operations. Future work may focus on creating more generalized models and integrating additional sensor data for multi-modal fault detection.

2.6 BLADE INSPECTION (Shakya, 2024)

The paper reviews recent advancements in the field of wind turbine blade inspection, focusing on the integration of drone technology and deep learning to enhance defect detection. Wind turbines are exposed to extreme environmental conditions, making their blades susceptible to defects like cracks, erosion, and delamination. Traditional inspection methods, such as manual inspections or rope-access inspections, are risky, time-consuming, costly, and often inaccurate. To overcome these limitations, the use of Unmanned Aerial Vehicles (UAVs), or drones, combined with Artificial Intelligence (AI) — particularly deep learning — is gaining significant traction. Drones offer flexible, safe, and efficient ways to capture high-resolution images and other data from difficult-to-access blade surfaces without the need to halt turbine operations (Shakya, 2024). The review highlights how deep learning algorithms, especially Convolutional Neural Networks (CNNs), are employed to automatically analyze drone-captured images. These AI models are trained to recognize various blade defects with high accuracy, reducing human error and enabling faster diagnosis. Some studies also explore the use of transfer learning, where pre-trained networks are adapted to specific inspection tasks, saving time and computational resources.

Key challenges identified include:

- The collection of large, diverse datasets is needed to train deep learning models effectively.
- Managing variations in lighting, weather conditions, and blade materials that can affect image quality and model performance.

- Ensuring real-time processing capabilities for immediate defect assessment during drone flights.

The paper also discusses emerging techniques such as:

- 3D mapping and photogrammetry for detailed blade surface modeling.
- The use of multi-modal data (combining visual, thermal, and LiDAR data) for more robust defect detection.
- Integration of autonomous navigation for drones, allowing them to plan optimized flight paths around turbines automatically.

Furthermore, the review touches on regulatory and practical issues like safety standards, data privacy, and maintenance scheduling based on predictive insights. In conclusion, integrating drones with deep learning offers transformative potential for the wind energy sector by enhancing blade inspection accuracy, reducing costs, and improving safety. However, continued research is needed to address technical challenges and to move towards fully autonomous, intelligent inspection systems.

2.7 ANOMALY DETECTION (Fakhry, 2025)

The research project titled "75MW AC PV Module Field Anomaly Detection Using Drone-Based IR Orthogonal Images with Res-CNN3 Detector", authored by Katleho Masita, Ali Hasan, and Thokozani Shongwe and published in IEEE in 2022, explores an innovative and efficient approach to monitoring and fault detection in large-scale photovoltaic (PV) systems (Fakhry, 2025). With the increasing deployment of solar power as a renewable energy source, maintaining the operational health of PV modules is crucial for maximizing energy output and minimizing downtime. This project addresses the challenge by employing drone-based infrared imaging combined with an advanced deep learning framework for automated and accurate anomaly detection. The methodology centers around the use of Res-CNN3, a deep convolutional neural network with 112 layers specifically designed for high-level feature extraction and classification of PV module defects. This architecture is particularly well-suited for handling complex image data and enables the system to detect a wide variety of faults with high precision. The model is trained using a substantial dataset of low-resolution Electro Luminescence (EL) images captured from multiple PV power plants situated in diverse climatic regions. This diverse training dataset allows the model to learn various fault patterns and improve its generalization ability across different environmental conditions and panel types. In addition to model training, the performance of the system is rigorously evaluated using standard machine learning metrics such as True Positives (TP), False Positives (FP), and Intersection over Union (IoU). These metrics provide insight into the detection accuracy and reliability of the system under field conditions. The results demonstrate that the model is capable of identifying multiple

types of PV module defects—such as cracks, hotspots, and soiling—with high accuracy and minimal false detections. Moreover, the system performs robustly even when processing low-resolution images, making it well-suited for real-time inspections where high-resolution equipment may not always be feasible. Among the key advantages of this approach is its efficient processing time, enabled by the optimized architecture and pre-trained learning techniques. The integration of drone technology enhances the scalability and coverage of inspection operations, significantly reducing the time, labor, and cost associated with traditional manual inspections. Additionally, the method's ability to analyze orthogonal IR images captured in the field introduces a non-invasive and safe alternative for defect identification. However, the study also discusses several limitations. One major challenge is the dependency on the quality and variability of the training dataset. If the training data does not sufficiently represent the variety of environmental and operational conditions encountered in the field, the model's generalization capability may be compromised. Another concern is the computational intensity of training and deploying such a deep neural network, which may require specialized hardware and expertise. Furthermore, adapting the model for different types of PV modules and field setups could introduce additional complexity. In summary, this project provides a compelling example of how modern AI techniques and aerial imaging technologies can be integrated to revolutionize maintenance and monitoring processes in the solar energy sector. By combining deep learning with drone-based inspection, it presents a scalable, accurate, and efficient solution for real-time anomaly detection in PV modules—paving the way for smarter, more sustainable energy infrastructure.

CHAPTER 3

METHODOLOGY / THEORY / MODELLING

3.1 APPROACH

3.1.1 Data Acquisition

- i. Objective:
 - To collect comprehensive vibration data from drone components for the purpose of assessing the drone's physical condition and facilitating fault detection through AI algorithms.
- ii. Data Sources:
 - Simulated Data: MATLAB simulations are used to generate controlled vibration data that mimics real-world fault scenarios in a repeatable environment.
 - Public Datasets: Additional vibration datasets are sourced from repositories like Kaggle, offering a diverse range of sensor data collected from drones under various conditions.
- iii. Data Collection Process:
 - Data is acquired from key drone components such as motors, propellers, and multirotor arms using accelerometer and gyroscope sensors.
 - Vibration signals are sampled over time and stored in a structured format (CSV, JSON, or SQL database) for further preprocessing and model training.
 - Time-series analysis is employed to extract meaningful features such as amplitude, frequency, and waveform patterns.
- iv. Significance:
 - The integrity and cleanliness of the dataset significantly affects the performance of machine learning models.
 - Well-structured data enables the development of robust fault detection algorithms capable of identifying early signs of mechanical issues like loose joints, damaged rotors, or arm cracks.

- Ensures the safety and reliability of drone flights, especially in critical missions like surveillance, delivery, and rescue operations.

v. Outcome:

- Vibration data becomes the foundational input for the AI model, enabling automated health diagnostics and predictive maintenance.

3.1.2 Fault Simulation and Data Calibration.

i. Objective:

- To simulate realistic fault conditions in drones and define vibration thresholds that help distinguish between safe and unsafe states.

ii. Simulation Process:

- Faults are deliberately introduced into drone arms by loosening screws, detaching and reattaching joints with adjustable plastic brackets, or partially unscrewing structural components.
- Multiple physical configurations are tested:

Fully Tightened (100%) – Represents the optimal and safe condition.

Half Tightened (50%) – Indicates minor structural degradation.

Slightly Tightened (10%) – Represents early-stage faults. Fully

Unscrewed – Indicates severe faults or potential failure.

iii. Data Collection & Calibration:

- For each condition, vibration data is collected using onboard sensors.
- Each dataset is labeled based on the fault condition and corresponding vibration amplitude.
- Vibration amplitudes are categorized into three distinct thresholds:

Low Amplitude (< 0.47 or 47,000 pulses): Safe State – Drone is structurally stable and ready for flight.

Medium Amplitude ($0.47 \leq x < 0.72$): Partially Safe State – Minor faults exist; inspection recommended before flight.

High Amplitude (≥ 0.72): Unsafe State – Drone is at risk; flight is not advisable.

iv. Significance:

- Helps in establishing baseline and abnormal patterns that can be used to train and test machine learning models.
- Enables the creation of a ground truth for the classification system in the monitoring platform.

3.1.3 Website for Real-Time Monitoring and Fault Detection

i. Objective:

- To provide an accessible, real-time web-based platform where users can monitor drone status and receive fault detection analysis based on vibration data input.

ii. Functionalities:

User Authentication:

- Secure login system where users must enter valid credentials.
- New users are required to register before gaining access to the system.

iii. Data Input Interface:

- Users can manually input or upload vibration data from the four drone arms.
- The interface allows real-time interaction with the fault detection engine.

iv. Fault Classification and Output:

- The backend AI model processes the input values.
- The system analyzes the data, classifies the fault status (safe, partially safe, unsafe), and displays whether the drone is ready for takeoff.

- v. User Experience:
 - Simple and intuitive UI for drone operators and technicians. Dashboard includes status indicators, previous test results, and recommendations for maintenance.

- vi. Purpose:
 - Empowers users to make informed decisions about drone usage based on real-time structural health data.
 - Reduces manual inspection efforts and enhances operational safety.

3.1.4 Testing and Validation

- i. Objective:
 - To rigorously evaluate the performance, accuracy, and robustness of the AI-based fault detection system across various conditions.
- ii. Dataset Utilization:
 - Vibration datasets from MATLAB simulations and Kaggle are used as test and validation sets.
 - Data is preprocessed and partitioned into training, validation, and testing sets for effective model assessment.
- iii. Model Validation Metrics:
 - Accuracy: Measures overall correctness of classification.
 - Precision and Recall: Assesses the model's ability to correctly identify fault types.
 - F1-Score: Balances precision and recall, especially important for imbalanced datasets.
 - Confusion Matrix: Visualizes the classification performance across all fault categories.
- iv. Testing Scenarios:

The model is tested against different fault conditions:

 - Normal operation (safe state)
 - Minor faults (partially safe)

Critical faults (unsafe state)

Realistic inputs are used to ensure the model can handle variability in sensor data.

v. Real-Time Testing via Website:

- The website interface is used to input live test data.
- Outputs are verified against expected results to ensure proper system functionality.

vi. Outcome:

- Validated AI model with high accuracy in detecting drone faults.
- System proven to be reliable and ready for deployment in real-world scenarios.

3.2 THEORETICAL FRAMEWORK

3.2.1 Random Forest

It is a supervised machine learning algorithm that belongs to the family of ensemble methods. It builds multiple decision trees during training and outputs the mode (majority vote) of the classes for classification problems or the mean prediction for regression Problems.

It is known for its high accuracy, robustness to noise, and ability to generalize well on unseen data, making it highly suitable for fault detection in mechanical systems like drones.

3.2.2 The Need for Random Forest in Drone Fault Detection

Drones are complex systems with many moving parts (e.g., motors, propellers, arms). Over time, these parts can develop faults due to wear and tear, impacts, or manufacturing defects. These faults can often be detected by changes in vibration patterns.

However, these changes:

- Can be subtle
- May vary due to flight dynamics or environmental conditions
- It might be non-linear and high-dimensional.

Random Forests are well-suited to handle these challenges by:

- Modeling complex decision boundaries,
- Handling large amounts of data and features,
- Reducing the risk of overfitting (unlike individual decision trees),
- Providing features of importance for interpretability.

3.2.3 Step-by-Step Workflow:

Step 1: Bootstrapping (Data Sampling)

- Random Forest starts by creating several bootstrap datasets from the original vibration dataset.
- Each bootstrap sample is generated by random sampling with replacement.
- This helps in creating a diverse set of data for training each tree.

Step 2: Decision Tree Construction

- For each bootstrap sample, a decision tree is trained.
- During tree construction, a random subset of features (e.g., RMS, peak-to-peak, skewness, FFT components) is selected at each split.
- The tree continues to split until it perfectly classifies the training data or meets stopping criteria (e.g., max depth, minimum samples).

Step 3: Voting (Ensemble Decision)

- After training, each tree provides a classification result.
- The final prediction is made by majority voting—the class (safe, partially safe, unsafe) most trees agree upon is chosen.

Mathematical Foundation

Let the dataset be:

$$D = \{(X1,Y1),(X2,Y2),...,(Xn,Yn)\} \quad \dots(1)$$

Where:

- feature vector from vibration signal (e.g., amplitude, frequency, RMS)
- label representing the fault state (0 = Safe, 1 = Partially Safe, 2 = Unsafe)

Random Forest builds:

- decision trees

For a new input , the predicted output is:

$$y = mode\{t1(x), t2(x), \dots, tT(x)\} \quad \dots(2)$$

Where "mode" indicates the class with the most votes.

3.2.4 Application to Vibration Fault Detection in Drones

Input Features (from sensors):

- Root Mean Square (RMS)
- Peak amplitude
- Crest factor
- Pulse count

Frequency-domain features:

- Dominant frequency (from FFT)
- Harmonics
- Band energy

Statistical features:

- Skewness
- Kurtosis

Output Labels:

- Class 0: Safe
Low vibration amplitude (<0.47)
- Class 1: Partially Safe
Medium amplitude (0.47–0.72)

- Class 2: Unsafe
High amplitude (≥ 0.72)

Training the Model:

- Labeled data is fed to Random Forest for training.
- The model learns which patterns of vibration correspond to which fault levels.

Real-Time Monitoring:

- When the drone is running, live vibration readings are input to the model.
- The model classifies the state of each arm and determines if the drone is ready to fly.

3.2.5 Conclusion

The Random Forest algorithm provides a robust, scalable, and interpretable framework for classifying vibration data in drones. Its ensemble nature ensures high accuracy, and its ability to handle high-dimensional and noisy data makes it ideal for this real-time safety-critical application.

3.3 TOOLS AND TECHNOLOGIES

3.3.1 Software

i Front-End Development:

- HTML (HyperText Markup Language): Structures the web pages and defines the content layout.
- CSS (Cascading Style Sheets): Enhances the visual appeal with styling, colors, and responsive design.
- JavaScript (JS): Adds interactivity, user input handling, and dynamic content updates.

ii Back-End Development:

- Python Django Framework:
 - Used for building the server-side application and handling user requests.
 - Provides a secure, scalable, and efficient environment for hosting the trained

AI model.

- Simplifies database interactions, authentication, and API development.

iii Database Management:

- SQL (Structured Query Language):
- Stores and manages user credentials, vibration data, and classification results.
- Ensures efficient data retrieval, storage, and security for real-time processing.

iv. AI Algorithm:

- Random Forest :
 - High Accuracy: Reduces overfitting compared to single decision trees.
 - Handles Noisy Data Well: Effective for real-world vibration datasets with variations.
 - Feature Importance: Identifies which vibration parameters are most critical for fault detection.
 - Model Training:
 - The dataset is split into training and testing sets to evaluate performance.
 - The model learns patterns from vibration data and classifies the drone's condition.
 - Performance Metrics:
 - Accuracy, Precision are used to validate predictions.
- Integration with Website:
- The trained Random Forest model is deployed on the Django backend for real-time classification.

3.4 DESIGN PROCESS

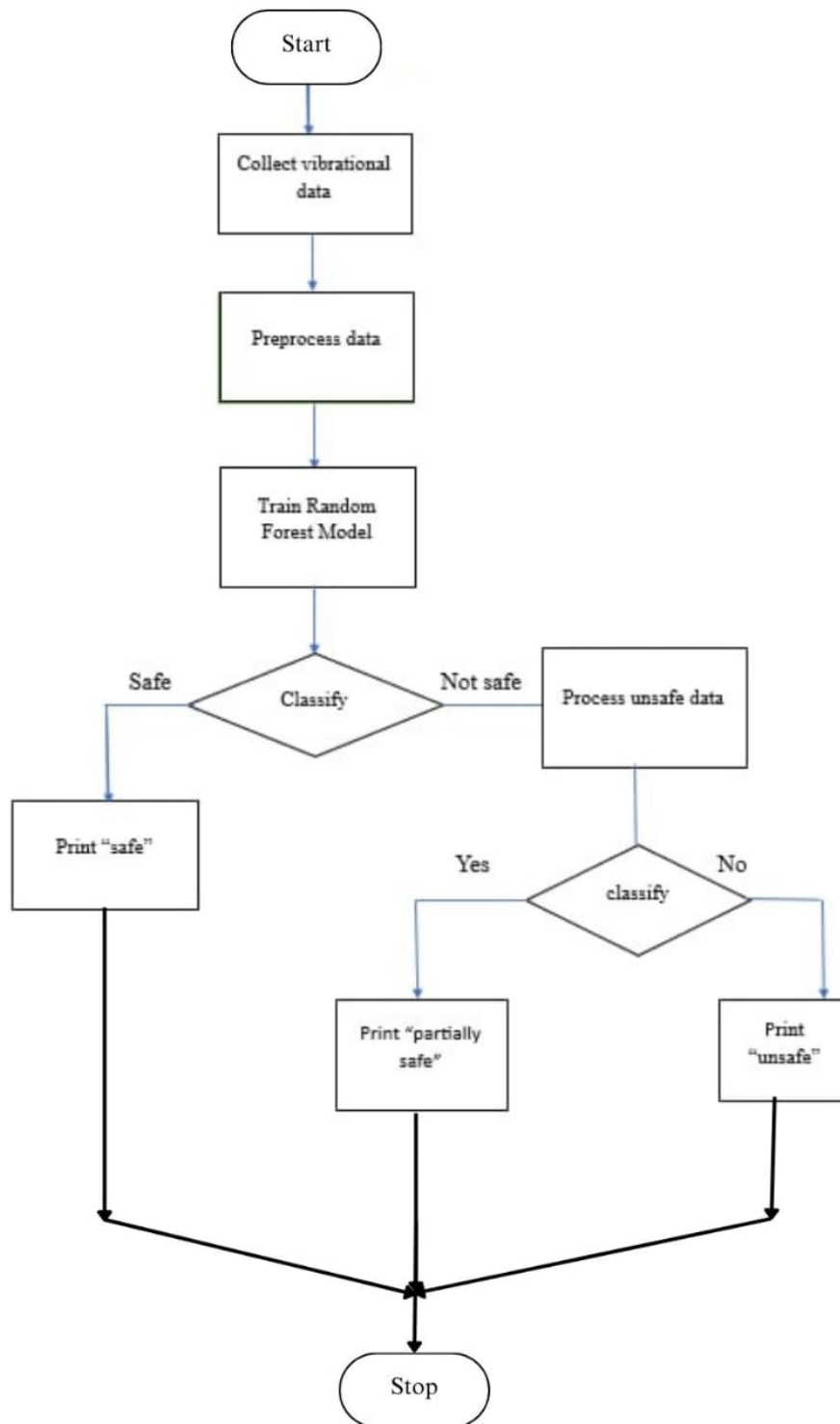


Fig 3.4 Flowchart

CHAPTER 4

WORK PLAN & TASK ALLOCATION

4.1 PROJECT TIMELINE

4.1.1 Phase 1: Planning And Requirements Gathering

Week 1 (Starting 09-01-2025):

- Conducted extensive research on the domain of drone malfunction detection.
- Identified and analyzed existing problems related to drone malfunctions.
- Proposed a viable solution to the most critical problem observed.
- Finalized the mini-project topic as “Intelligent Vibration Analysis for Drone Malfunction Detection.”
- Gained a fundamental understanding of drone architecture and operation.
- Drafted the project abstract and collected relevant base papers from IEEE Xplore.
- Selected the base paper titled “Vibration Based Fault Detection in Drones Using AI” by Mohammed Hazwan Mohd Ghazali and Wan Rahiman, Member, IEEE.
- Conducted a detailed study of the base paper and reviewed related literature.
- Created and presented the zeroth presentation along with a supporting PowerPoint.

Week 2:

- Deepened understanding of the selected base paper and its hardware implications.
- Identified and documented the software and hardware requirements for the project.

- Formulated a comprehensive overview of the project development strategy.
- Studied the fundamentals of machine learning algorithms.
- Initiated learning of HTML and CSS for the web development part of the project.
- Began studying Python programming and grasped core concepts of AI models.
- Learned the working and implementation of Linear Regression and Random Forest algorithms.
- Collected necessary details and began documentation work.
- Successfully completed the Software Requirements Specification (SRS) and Software Design Document (SDD).

4.1.2 Phase 2: Development (Week 3 to Week 8)

Week 3:

- Explored the core functioning of the Linear Regression algorithm in detail.
- Collected vibration datasets from MATLAB and Kaggle.
- Analyzed both datasets and selected the MATLAB dataset due to its better relevance and quality.

Week 4:

- Implemented the Linear Regression model using the selected dataset.
- Trained and evaluated the model, which resulted in an accuracy of only 47%, prompting reconsideration of the algorithm.

Week 5:

- Shifted focus to the Random Forest algorithm due to its higher expected performance.

- Studied its working principles, use cases, and potential benefits.

Week 6:

- Successfully developed the machine learning model using the Random Forest algorithm.
- Achieved an improved accuracy of 90% during initial evaluation.

Week 7:

- Designed and developed the front-end of the system using HTML, CSS, and JavaScript.

Week 8:

- Built the back-end using Python and Django framework, integrating it with the trained AI model.

4.1.2 Phase 3: Testing and Deployment (Week 9 to Week 10)

Week 9:

- Conducted comprehensive testing of the system.
- Identified and resolved bugs and errors in the codebase to improve functionality and stability.

4.2 TASK ALLOCATION

Table 4.2 Task Allocation

SL NO	ROLES	ASSIGNED MEMBER
1.	Research and Analysis	Hamna And Greeshma
2.	Front End Development	Greeshma And Abdulla
3.	Back End Development	Adhil B
4.	Data Base Manager	Hamna
5.	Testing And Debugging	Adhil B And Abdulla
6.	Documentation And Reporting	Hamna And Greeshma

4.3 COLLABORATION TOOLS

In this project, “Intelligent Vibration Analysis for Drone Malfunction Detection”, Google Colaboratory (Colab) was used as a key collaboration and development tool. Google Colab is a cloud-based interactive coding environment that supports Python programming and is especially useful for data science and machine learning tasks. It allowed our team to collaboratively develop, test, and run our machine learning models without the need for powerful local hardware.

The platform provided access to essential Python libraries such as Pandas, Matplotlib which were crucial for data preprocessing, visualization, and model implementation. We used Google Colab to train and evaluate our Random Forest classifier using drone vibration data, sourced from simulations and public datasets.

CHAPTER 5

EXPERIMENTAL DESIGN AND CODING

5.1 EXPERIMENTAL SETUP

5.1.1 Dataset

- A publicly available drone vibration dataset was used for training and testing.
- The dataset included labeled samples representing different conditions: normal, caution, and danger states of drone operation.

5.1.2 Software and Tools

- Google Colab: Used as the development and execution environment for the project.
- Python : Primary programming language used for implementation.
- Libraries Used:
 - Pandas – for data handling and preprocessing.
 - Matplotlib, Seaborn – for data visualization.
 - Scikit-learn – for implementing machine learning models (Random Forest Classifier).

5.1.3 Machine Learning Model

- A Random Forest Classifier was trained using the preprocessed vibration dataset.

ZA

- The model was evaluated using accuracy and confusion matrix metrics.

5.1.4 Simulated Environment

- Since real drone hardware was not accessible, the malfunction detection was validated through simulated data inputs.
- The outputs were analyzed to identify patterns corresponding to safe, caution, and dangerous drone states.

5.2 CODING DETAILS:

5.2.1 Development of Django Framework

```
from django.shortcuts import  
  
renderredirect  
from django.contrib.auth import authenticate, login as auth_login, logout as auth_logout  
from django.contrib.auth.models import User  
from django.http import JsonResponse  
from django.contrib import messages  
from django.views.decorators.csrf import csrf_exempt  
import os  
import joblib  
from .forms import SensorInputForm  
  
# Load trained model dynamically  
  
model_path = os.path.join(os.path.dirname(_file_), "models", "random_forest_model.pkl")  
model = joblib.load(model_path)  
  
# Predict Drone Fault  
  
@csrf_exempt  
def predict_fault(request):  
    prediction = None  
    safety_status = None  
  
    if request.method == "POST":
```

```

form = SensorInputForm(request.POST)
if form.is_valid():
    # Extract sensor values
    sensor_values = [
        form.cleaned_data["sensor_a"],
        form.cleaned_data["sensor_b"],
        form.cleaned_data["sensor_c"],
        form.cleaned_data["sensor_d"],
    ]
    # Make prediction

prediction = model.predict([sensor_values])[0]

    # Determine safety status

    if prediction < 0.4:
        safety_status = "■ Safe"
    elif 0.4 <= prediction < 0.65:
        safety_status = "⚠ Partially Safe"
    else:
        safety_status = "✖ Not Safe"
else:
    form = SensorInputForm()

    return render(request, "predict_form.html", {"form": form, "prediction": prediction,
"safety_status": safety_status})

# Render Register Page

def register_page(request):
    return render(request, "register.html")

# Render Login Page

def login_page(request):
    return render(request, "login.html")

# Register API

@csrf_exempt
def register(request):
    if request.method == "POST":

```

try:

```
username = request.POST.get("username")
email = request.POST.get("email")
password = request.POST.get("password")
password2 = request.POST.get("password2")
if password != password2:
    messages.error(request, "Passwords do not match.")
    return redirect('register_page')

if User.objects.filter(username=username).exists():
    messages.error(request, "Username already taken.")
    return redirect('register_page')

User.objects.create_user(username=username, email=email, password=password)
messages.error(request, "User registered Succesfully")
return redirect('login_page')
except Exception as e:
    return JsonResponse({"error": str(e)}, status=500)
```

```
return JsonResponse({"error": "Invalid request method. Use POST."}, status=400)
```

Login API

@csrf_exempt

def login(request):

if request.method == "POST":

try:

username = request.POST.get("username")

password = request.POST.get("password")

user = authenticate(username=username, password=password)

if user:

auth_login(request, user)

return redirect('predict_form')

else:

messages.error(request, "Invalid username or password.")

return redirect('login_page')

except Exception as e:

return JsonResponse({"error": str(e)}, status=500)

```
return JsonResponse({"error": "Invalid request method. Use POST."}, status=400)
```

Logout API

```
def logout(request):  
    auth_logout(request)  
    return redirect('login_page')
```

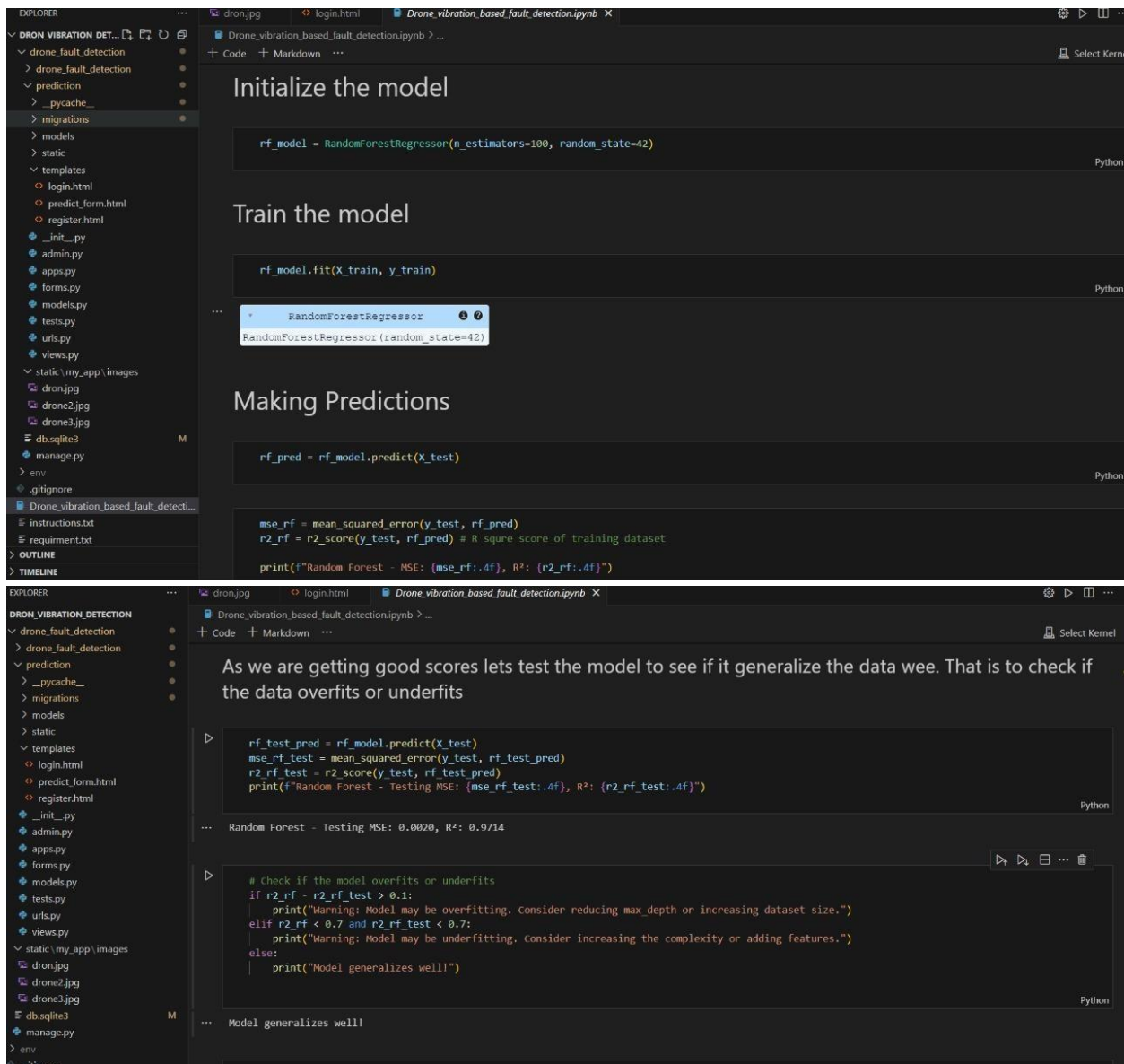


Fig 5.2 Initializing and Training the Model

5.3 CLASS DIAGRAM

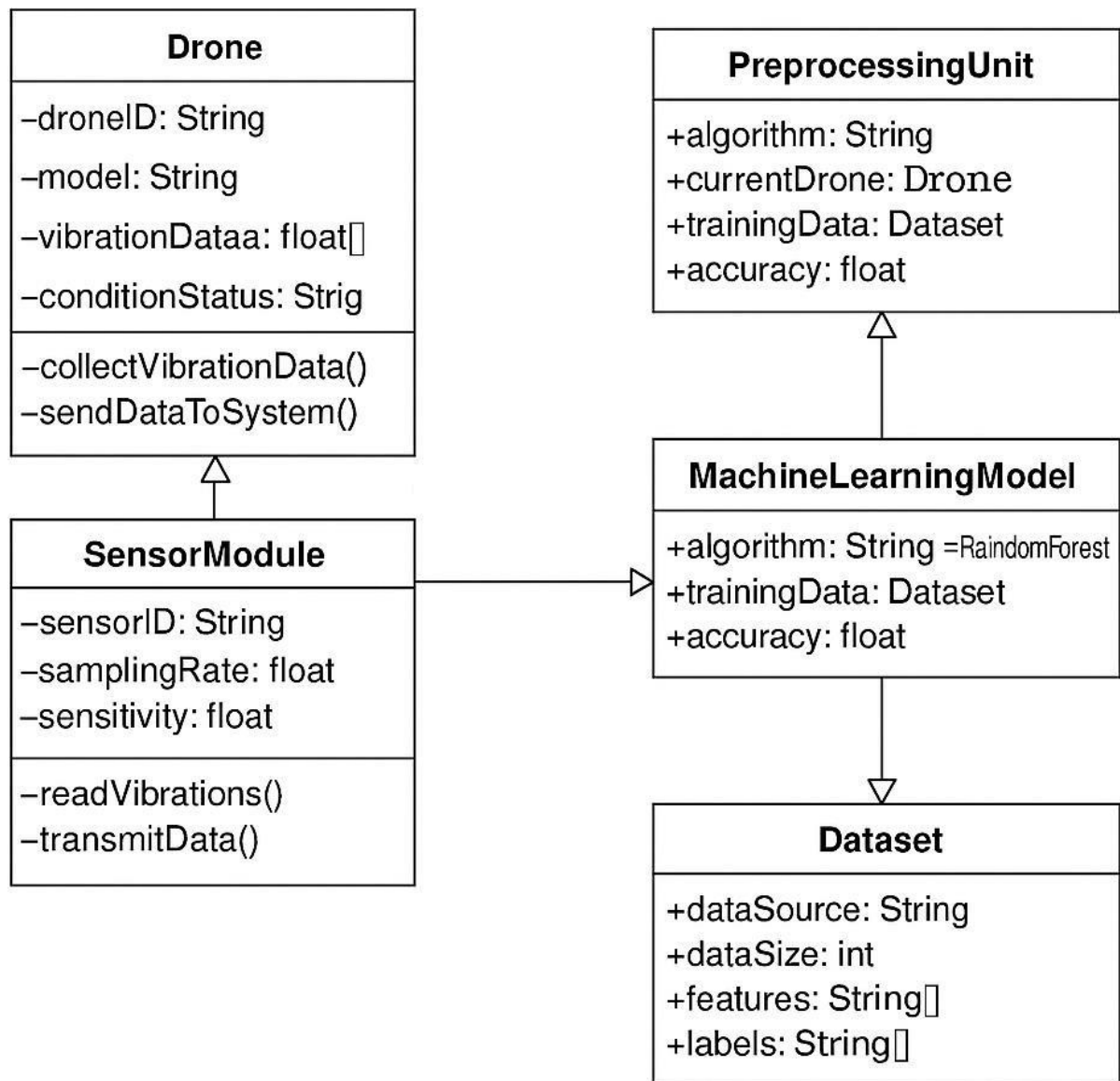


Fig 5.3 Class Diagram Of the Intelligent Vibration Analysis System for Drone Malfunction Detection

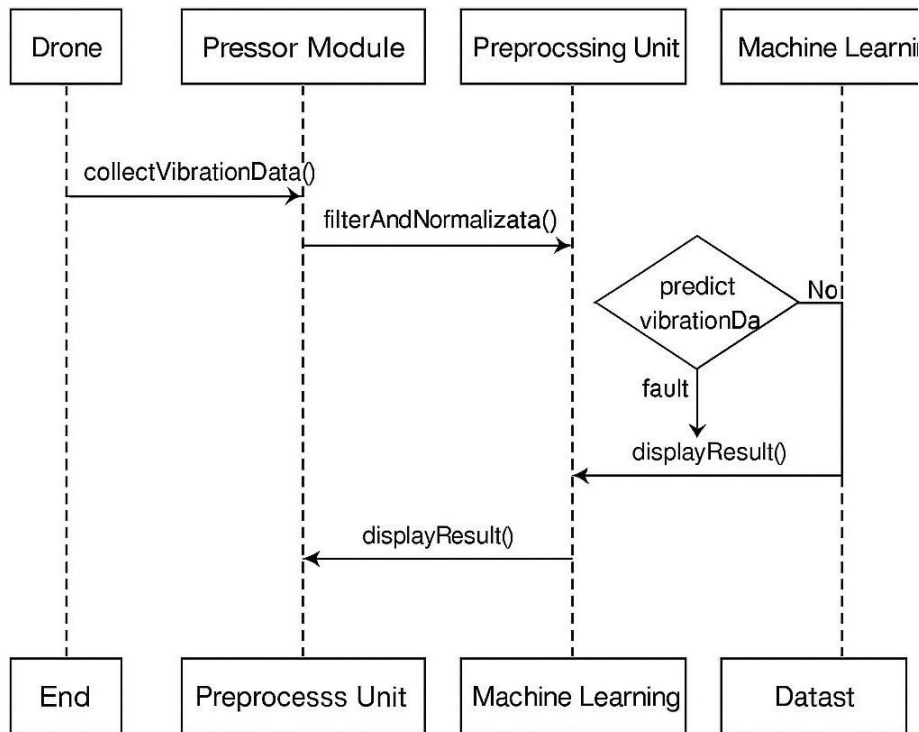


Fig 5.4 Sequence Diagram

5.4 TESTING AND VALIDATION

For the purpose of testing and validation, a dataset comprising 1000 numerical vibration data values from the drone was collected. The dataset was split into two subsets: the first 600 values were used to train the Random Forest algorithm, while the remaining 400 values were reserved for testing the model's performance.

To evaluate the effectiveness of the trained model, the R^2 score (coefficient of determination) was used as the performance metric. The model achieved a high accuracy of 97%, indicating a strong predictive capability in classifying the drone's condition based on the vibration data.

CHAPTER 6

RESULTS AND CONCLUSION

6.1 ANALYSIS

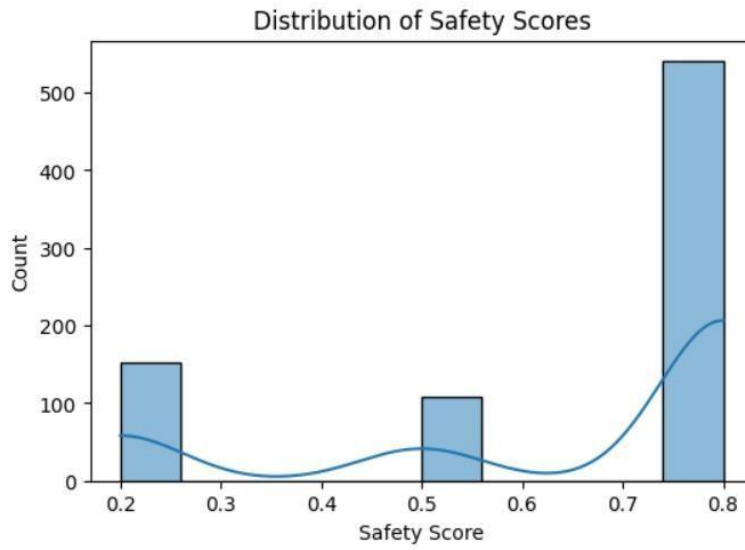


Fig 6.1 shows how data is distributed

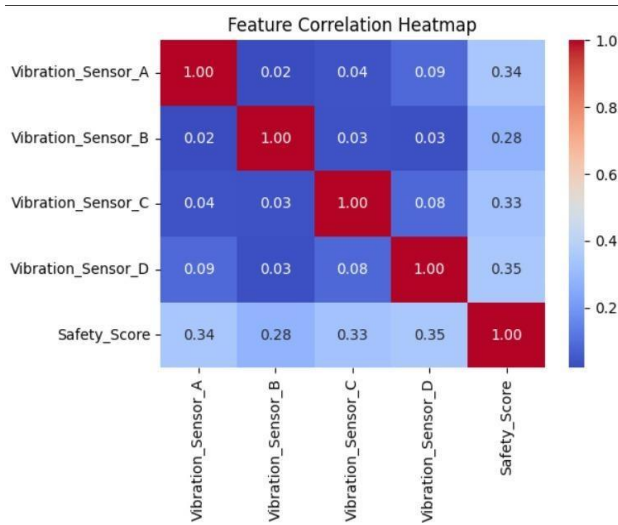


Fig 6.2 Heat Map

6.2 RESULT

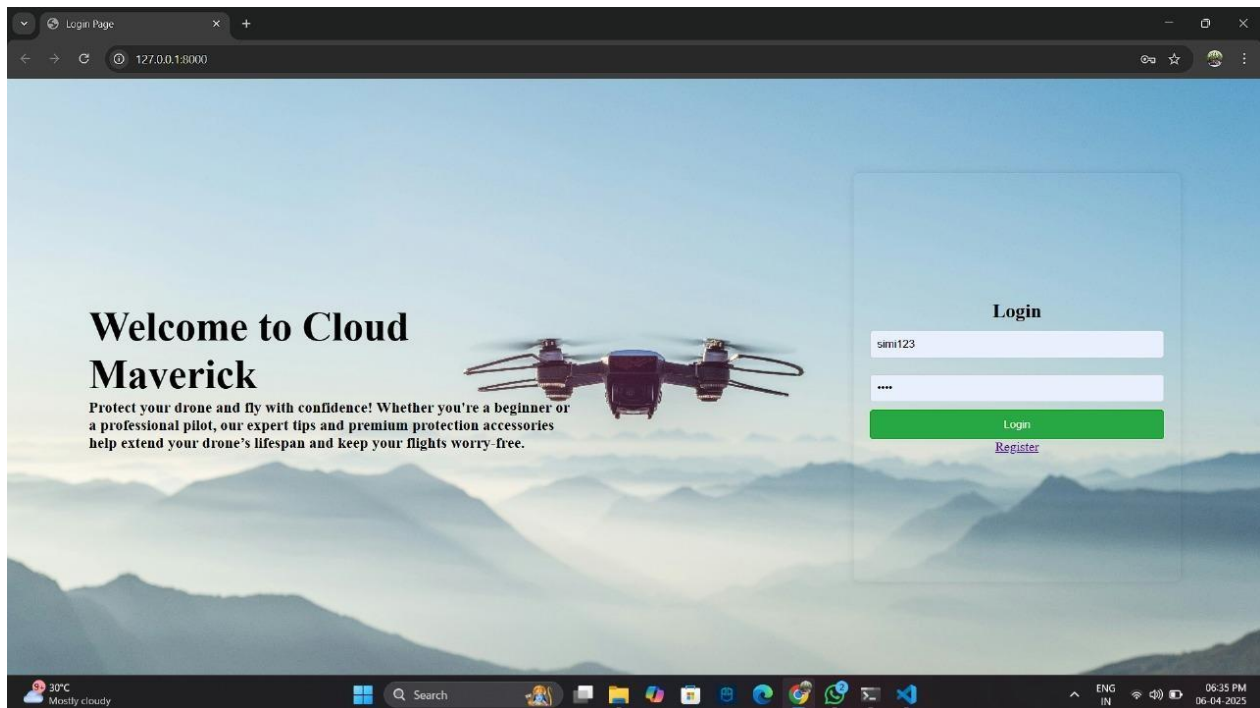


Fig 6.3 The front page of the website

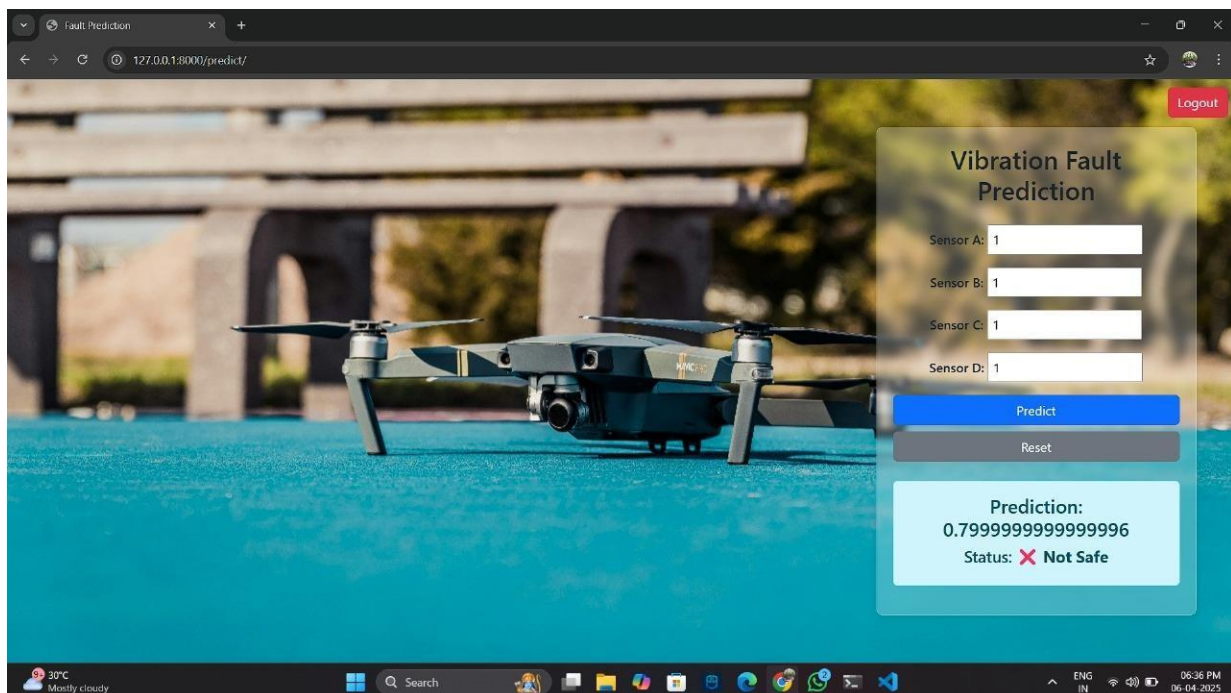


Fig 6.4 The drone is unsafe for takeoff

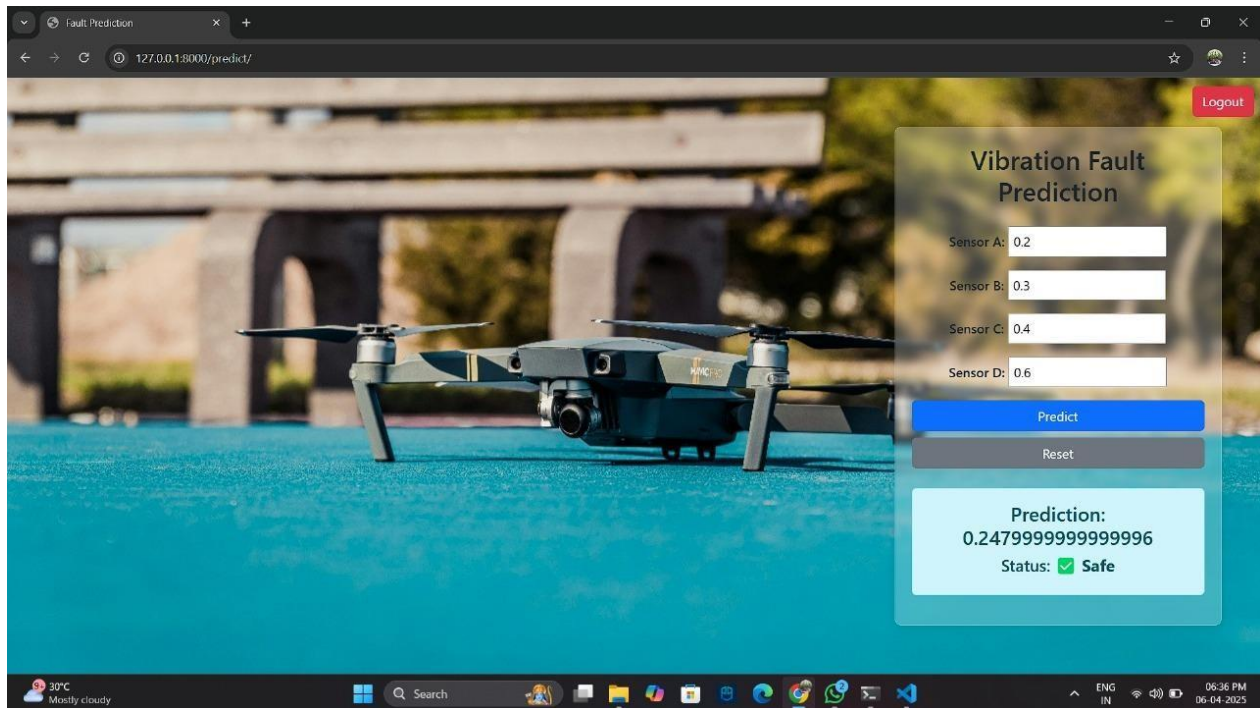


Fig 6.5 The drone is safe for takeoff

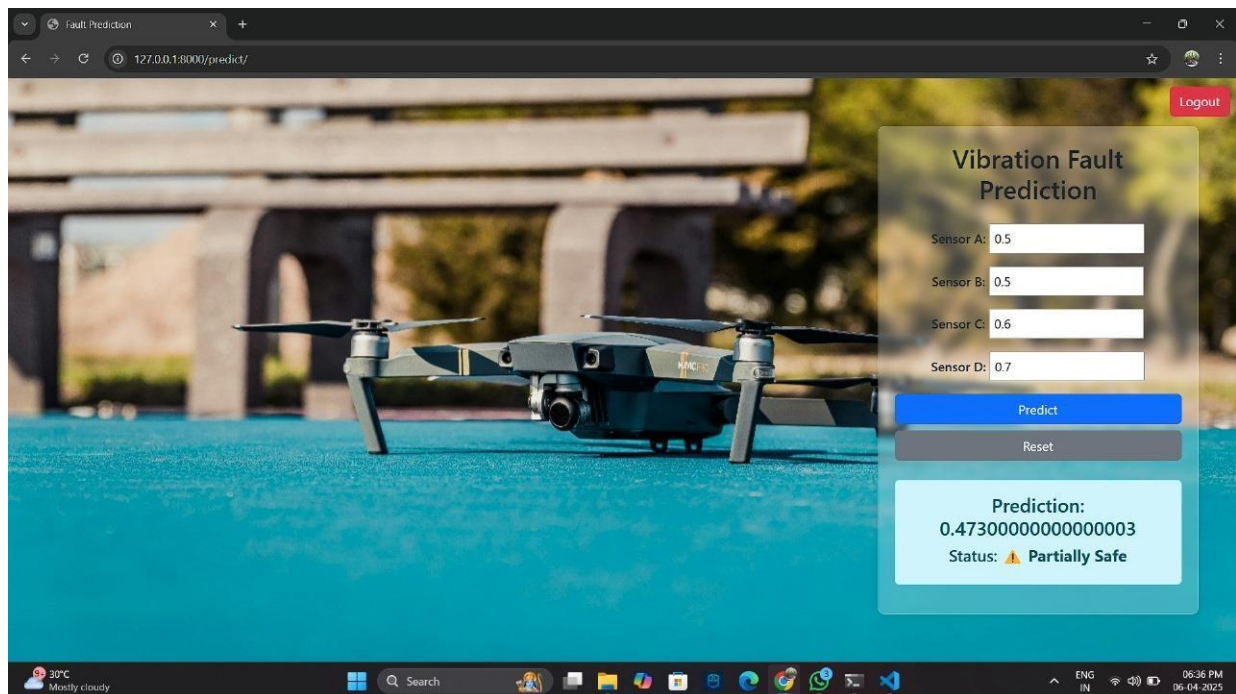


Fig 6.6 The drone is Partially safe for takeoff

Importing Libraries

```
import pandas as pd # Used for data manipulation and loading datasets (Excel, CSV)
import matplotlib.pyplot as plt # A plotting library for visualizing data (e.g., histograms, scatter plots).
import seaborn as sns # Built on matplotlib, used for advanced data visualization with better aesthetics.
from sklearn.model_selection import train_test_split # Splits the dataset into training and testing sets.
from sklearn.linear_model import LinearRegression # Implements Linear Regression for predictive
from sklearn.ensemble import RandomForestRegressor # Implements Random Forest Regression for predictive
from sklearn.metrics import mean_squared_error # Measures prediction error (how far predictions are from actual values).
from sklearn.metrics import r2_score # Evaluates model performance (how well it explains variability in data).
```

Python

Load Dataset

```
file_path = "/content/NeuroFuzzy_1000_Dataset.xlsx"
training_data = pd.read_excel(file_path, sheet_name="Training")
testing_data = pd.read_excel(file_path, sheet_name="Testing")
```

Python

Fig 6.7 Importing Libraries

Exploratory Data Analysis

✓ Checking the first and last 5 rows of data

```
training_data.head()
```

...

	Input 1	Input 2	Input 3	Input 4	Output
0	0.30	0.25	0.75	0.50	0.8
1	0.15	0.24	0.30	0.40	0.2
2	0.70	0.70	0.80	0.85	0.8
3	0.60	0.50	0.55	0.45	0.2
4	0.35	0.20	0.70	0.80	0.8

```
testing_data.head()
```

...

	Input 1	Input 2	Input 3	Input 4	Output
0	0.117037	0.146515	0.503840	0.430278	0.2
1	0.814682	0.631141	0.612810	0.693753	0.8
2	0.324855	0.859320	0.819422	0.945213	0.8
3	0.246228	0.974222	0.531889	0.784233	0.8
4	0.342713	0.570838	0.202075	0.705572	0.5

Fig 6.8 Data Preprocessing

<code>training_data.tail()</code>					
...	Input 1	Input 2	Input 3	Input 4	Output
795	0.877799	0.021556	0.933502	0.450394	0.8
796	0.582433	0.559841	0.668464	0.205672	0.2
797	0.070684	0.300819	0.206776	0.899651	0.8
798	0.922745	0.939410	0.653851	0.762586	0.8
799	0.800372	0.980904	0.072052	0.882486	0.8
<code>testing_data.tail()</code>					
...	Input 1	Input 2	Input 3	Input 4	Output
95	0.576053	0.850357	0.908102	0.992359	0.8
96	0.810628	0.911647	0.552175	0.802262	0.8
97	0.403843	0.639276	0.032940	0.424227	0.2
98	0.988439	0.255370	0.053863	0.728864	0.8
99	0.089999	0.088666	0.805063	0.498354	0.8

Fig 6.9 Training and Testing with datasets

Statistical Analysis + Code + Markdown					
<code>training_data.describe()</code>					
...	Vibration_Sensor_A	Vibration_Sensor_B	Vibration_Sensor_C	Vibration_Sensor_D	Safety_Score
count	800.000000	800.000000	800.000000	800.000000	800.000000
mean	0.505643	0.517003	0.501960	0.505170	0.645500
std	0.292595	0.280698	0.274645	0.279527	0.238224
min	0.000000	0.000000	0.000000	0.000000	0.200000
25%	0.235720	0.284208	0.267388	0.246533	0.500000
50%	0.524689	0.526188	0.500000	0.514119	0.800000
75%	0.755030	0.754552	0.725858	0.741103	0.800000
max	0.999900	0.999329	0.999100	0.997560	0.800000
<code>training_data.isna().sum()</code>					
...	0				
Vibration_Sensor_A	0				
Vibration_Sensor_B	0				
Vibration_Sensor_C	0				
Vibration_Sensor_D	0				
Safety_Score	0				

Fig 6.91 Training and Testing with datasets

Data Manipulation

Renaming column names for clarity

```
training_data.rename(columns={
    "Input 1": "Vibration_Sensor_A",
    "Input 2": "Vibration_Sensor_B",
    "Input 3": "Vibration_Sensor_C",
    "Input 4": "Vibration_Sensor_D",
    "Output": "Safety_Score"
}, inplace=True)

testing_data.rename(columns={
    "Input 1": "Vibration_Sensor_A",
    "Input 2": "Vibration_Sensor_B",
    "Input 3": "Vibration_Sensor_C",
    "Input 4": "Vibration_Sensor_D",
    "Output": "Safety_Score"
}, inplace=True)
```

Fig 6.92 Manipulation of Data

6.3 CONCLUSION

- Objective Achieved: Developed an intelligent drone fault detection system.
- Effective Use of Vibration Data: Enabled early detection of malfunctions.
- Machine Learning Integration: Random Forest improved safety and reliability.
- User Interface: Real-time website monitoring made it practical and user-friendly.
- Alignment with Goals: The project fulfilled its aim of proactive drone maintenance.

6.4 LIMITATIONS

- Limited Testing Environment: No outdoor or real-flight testing was done.
- Basic Fault Simulation: Only simple mechanical faults were simulated; not all real issues covered.
- Sensor Setup: External sensors add weight and complexity to the drone.
- Model Limitation: Only Random Forest was used; other models like deep learning weren't explored.

CHAPTER 7

FUTURE SCOPE

7.1 POTENTIAL IMPROVEMENTS

- **Real-time Monitoring:** Implementing real-time data streaming and live monitoring dashboards would allow users to view drone health status dynamically during flight.
- **Multi-Model Comparison:** Integrating and comparing different machine learning models (e.g., SVM, XGBoost, LSTM) could help identify the most efficient and accurate model for specific use cases.
- **User Interface Enhancements:** Improving the frontend with data visualizations and flight status summaries can make the tool more user-friendly and accessible to non-technical users.

7.2 APPLICATIONS

- **Drone Maintenance:** This system can be used by drone service providers to detect faults early, reducing downtime and maintenance costs.
- **Agriculture:** In precision farming, drones are vital; ensuring their reliability through vibration analysis helps improve productivity and safety.
- **Disaster Management:** Drones used in rescue operations can benefit from real-time health monitoring to prevent mid-mission failures.
- **Logistics and Delivery:** Companies like Amazon and Zipline can integrate this system into delivery drones to ensure package safety and avoid accidents.
- **Military Surveillance:** Enhancing drone reliability is critical in defense applications, where failures could compromise missions.

7.3 RESEARCH DIRECTIONS

- **Deep Learning Integration:** Future work could explore deep learning techniques like CNNs or RNNs for improved fault classification accuracy using raw or time-series data.

- Environmental Factors Study: Extending research to examine how external factors like wind, temperature, or altitude affect vibration data and drone performance.
- Scalability Testing: Investigate how the system performs across multiple drone types and under different payload conditions for broader industry adoption.

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